

WIGHT ST CATHERINES POINT, United Kingdom

WMO: 038660

Lat: 50.5756N

Lon: 1.2969W

Elev: 79

StdP: 14.65

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.6	32.9	22.0	16.6	34.4	24.8	19.0	35.8	42.2	48.1	37.4	48.2	11.2	50	0.686

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	73.2	64.6	70.6	63.2	68.3	62.1	65.7	71.1	64.5	68.6	63.4	66.8	12.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.8	88.7	67.6	62.7	85.6	66.3	61.8	82.8	65.4	30.6	71.5	29.6	68.8	28.8	67.0	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
35.5	31.6	28.1	DB	28.8	77.7	3.5	3.0	26.2	79.9	24.2	81.7	22.2	83.4	19.7	85.6	
			WB	26.8	68.1	3.3	1.5	24.4	69.2	22.5	70.0	20.6	70.8	18.2	71.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.8	44.0	43.5	45.5	48.9	53.8	58.6	62.5	63.3	60.7	56.2	50.2	45.8
	DBStd	8.20	4.98	4.51	3.94	4.20	4.16	3.38	3.33	2.89	3.20	4.12	5.01	5.52
	HDD50	798	187	182	143	65	9	0	0	0	0	7	58	147
	HDD65	4491	650	602	604	482	347	197	94	70	133	272	444	596
	CDD50	1825	2	1	4	33	128	257	388	411	321	200	64	16
	CDD65	42	0	0	0	0	1	4	17	16	4	0	0	0
	CDH74	56	0	0	0	0	1	6	26	21	2	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
(14) Wind	WSAvg	13.4	15.1	13.5	12.9	11.3	12.2	12.2	12.5	12.9	13.4	14.6	14.4	15.4
(15) Precipitation	PrecAvg	30.5	3.5	2.5	1.9	2.0	1.8	1.9	1.7	2.0	2.3	3.6	3.8	3.5
	PrecMax	40.7	7.4	5.9	5.4	4.8	4.9	5.0	4.4	5.5	5.2	6.7	7.6	7.0
	PrecMin	23.3	0.6	0.2	0.2	0.0	0.3	0.2	0.1	0.2	0.2	1.0	1.0	1.0
	PrecStd	4.9	1.7	1.6	1.0	1.2	1.1	1.2	1.1	1.3	1.3	1.5	1.9	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.2	52.7	57.2	65.0	70.2	74.1	77.3	76.7	72.4	66.4	60.7	56.4
		MCWB	51.2	49.1	50.4	56.0	60.5	64.6	65.9	66.8	65.1	61.8	58.2	54.2
	2%	DB	52.1	51.1	53.8	60.2	65.6	69.3	73.2	72.9	69.1	63.9	58.9	55.2
		MCWB	50.3	48.7	49.0	53.4	57.9	61.8	64.1	65.2	63.1	60.3	56.5	53.2
	5%	DB	51.2	50.0	51.7	57.1	62.6	66.3	70.4	70.1	66.9	62.6	57.8	54.1
		MCWB	49.4	48.1	48.0	51.3	56.6	59.9	62.9	63.5	61.6	59.6	55.4	52.0
	10%	DB	50.2	49.1	50.3	54.6	60.0	63.9	67.9	67.8	65.3	61.5	56.8	52.9
		MCWB	48.3	47.2	47.5	50.1	55.1	58.6	62.0	62.2	60.7	58.6	54.2	50.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	50.5	51.6	57.2	62.3	65.5	67.3	68.0	66.1	62.9	59.3	55.0
		MCDB	52.6	51.2	54.6	63.2	68.0	72.3	74.6	74.4	70.7	65.1	60.0	55.5
	2%	WB	50.9	49.5	50.2	54.3	59.2	62.9	65.2	66.1	64.3	61.4	57.4	53.8
		MCDB	51.6	50.4	52.2	59.1	63.7	68.3	70.9	71.6	67.4	63.1	58.3	54.7
	5%	WB	49.9	48.6	49.2	52.2	57.4	61.0	63.8	64.7	63.0	60.3	56.0	52.6
		MCDB	50.8	49.6	50.9	55.6	61.2	65.0	68.5	68.4	65.5	62.0	57.4	53.7
	10%	WB	48.8	47.7	48.3	50.8	55.8	59.5	62.7	63.5	62.0	59.3	54.7	51.1
		MCDB	50.0	48.8	49.8	53.5	59.1	62.6	66.6	66.5	64.4	61.2	56.5	52.5
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.1	7.0	8.1	8.1	7.8	7.4	7.6	8.0	6.8	6.4	6.0
		MCDBR	5.2	5.8	8.3	11.4	12.0	12.2	12.1	10.9	9.9	6.6	5.7	4.9
	5% WB	MCWBR	6.1	5.4	6.6	8.0	8.4	8.4	7.9	7.5	7.2	5.6	5.9	6.0
		MCWBR	5.1	5.2	6.9	9.3	10.6	10.4	9.9	9.0	7.9	5.8	5.2	4.7
(40) Clear-Sky Solar Irradiance	taub		0.334	0.350	0.387	0.403	0.406	0.402	0.395	0.389	0.378	0.364	0.337	0.325
		taud	2.383	2.392	2.291	2.278	2.319	2.358	2.384	2.418	2.435	2.443	2.429	2.367
	Ebn at Noon		218	247	256	266	270	272	272	268	258	239	216	201
		Edh at Noon	21	26	34	38	38	37	36	33	29	24	19	18
(44) All-Sky Solar Radiation	RadAvg		264	498	892	1428	1718	1902	1806	1515	1115	626	324	207
	RadStd		19	58	111	125	101	172	133	111	80	49	36	24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page